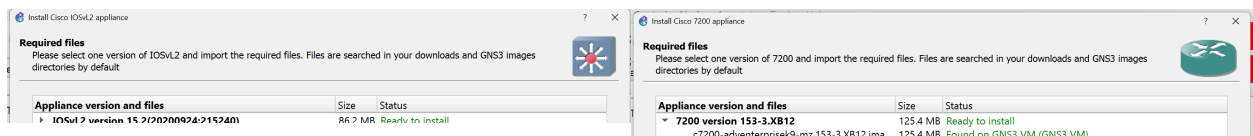


3 - STP, VLAN, FT

Name of report: INR_NetEng_LAB_3_Nikita_Niakhai

Preparation

- I chosen CISCO 7200 as my routing solution and CISCO IOSvL2 as a switching solution - one vendor for all type of net connectors.



Figures.

- Created the project.
- Increased memory and CPUs for my VM to 10696MB.
- Created a snapshot "Before".

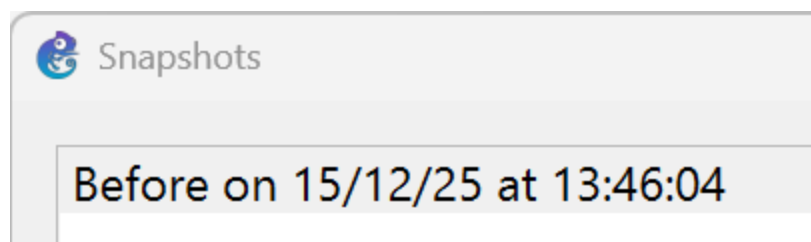


Figure. Snapshot is done

VLANs

1. Setup

- Setup topology, chosen for all nodes VM. Then added my routing and switching solutions from Cisco, because to provide communication they must be from on vendor..

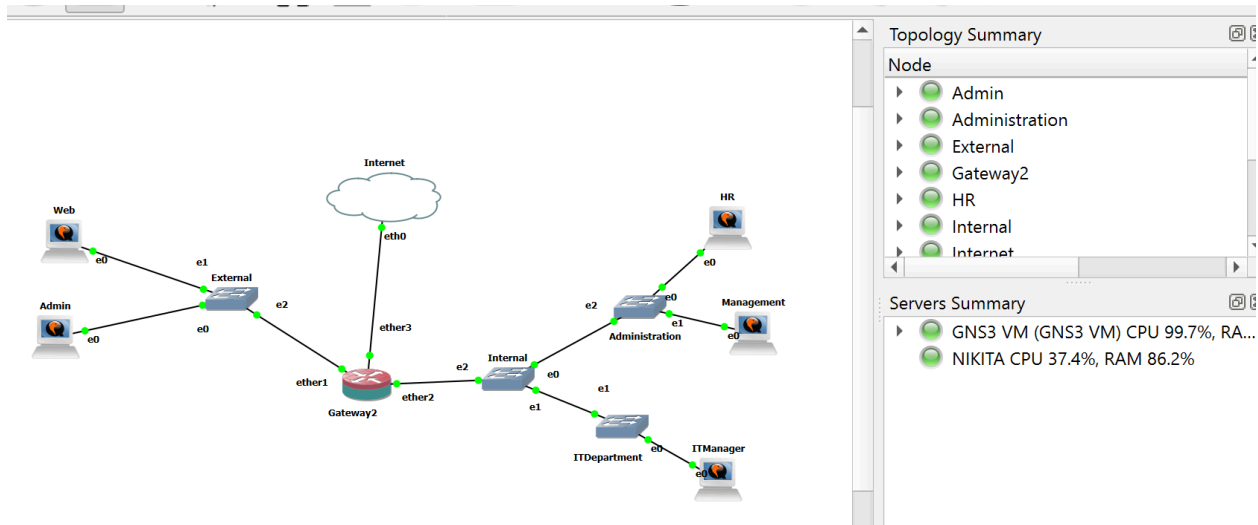


Figure. Topology

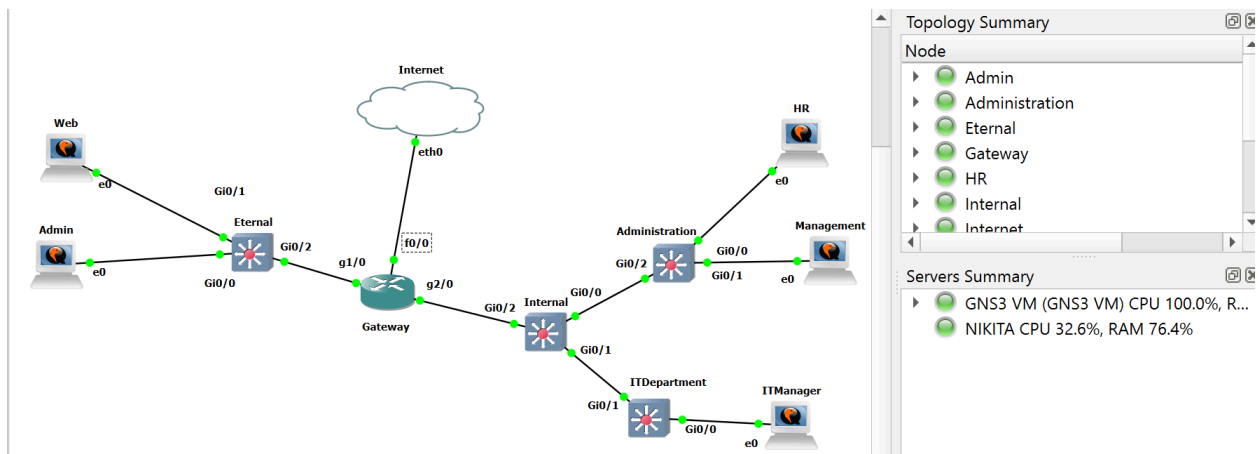
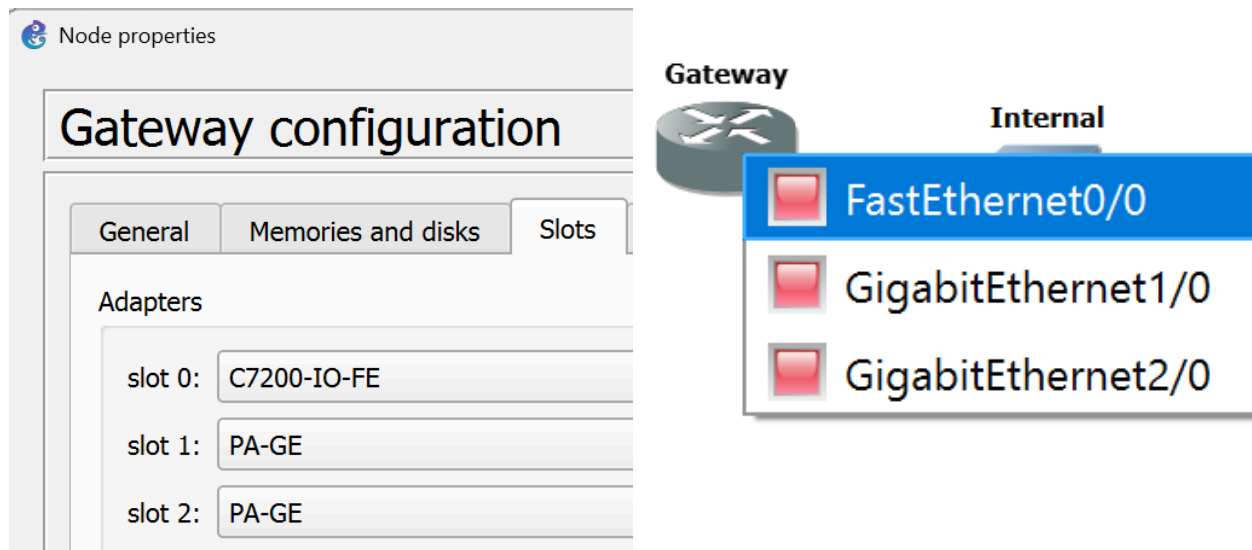


Figure. Topology

- I created new interfaces for gateway. FE - for the internet, GE - for switches.



Figures

2. Created router configuration

- enabled, entered configuration mode
- setup dhcp-client for internet access (ether1)
- setup static net masks for internal and external networks
- showed ip table on

```
Gateway#enable
Gateway#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Gateway(config)#interface f0/0
Gateway(config-if)#ip address dhcp
Gateway(config-if)#no shut
Gateway(config-if)#interface
*Dec 15 12:40:36.339: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Dec 15 12:40:37.339: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Gateway(config-if)#interface
*Dec 15 12:40:45.435: %DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned DHCP address 192.168.203.131, mask 255.255.255.0, hostname Gateway
```

Figure

```

Gateway(config-if)#interface g1/0
Gateway(config-if)#ip address 192.168.1.1 255.255.255.0
Gateway(config-if)#no shut
Gateway(config-if)#interface g2/0
Gateway(config-if)#ip address 10.0.0.1/24 255.255.255.0
^
% Invalid input detected at '^' marker.

Gateway(config-if)#ip address 10.0.0.1 255.255.255.0
Gateway(config-if)#exit

```

Figure

```
Gateway(config)#exit
```

Figure

```

Gateway#show ip inter br

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.203.131	YES	DHCP	up	up
GigabitEthernet1/0	192.168.1.1	YES	manual	up	up
GigabitEthernet2/0	10.0.0.1	YES	manual	administratively down	down

Figure. One cable is down, so we need to UP it:

```

Gateway(config)#interface g2/0

Gateway(config-if)#no shutdown
Gateway(config-if)#exit
Gateway(config)#e
*Dec 15 12:46:57.979: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Dec 15 12:46:58.979: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
Gateway(config)#exit
Gateway#
*Dec 15 12:47:02.135: %SYS-5-CONFIG_I: Configured from console by console
Gateway#show ip inter br

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.203.131	YES	DHCP	up	up
GigabitEthernet1/0	192.168.1.1	YES	manual	up	up
GigabitEthernet2/0	10.0.0.1	YES	manual	up	up

Figure

3. Configured hosts to use static IPs

```

# Web
sudo ip a add 192.168.1.2/24 dev ens3

```

```
sudo ip route add default via 192.168.1.1 dev ens3
```

Admin

```
sudo ip a add 192.168.1.3/24 dev ens3
```

```
sudo ip route add default via 192.168.1.1 dev ens3
```

HR

```
sudo ip a add 10.0.0.2/24 dev ens3
```

```
sudo ip route add default via 10.0.0.1 dev ens3
```

Management

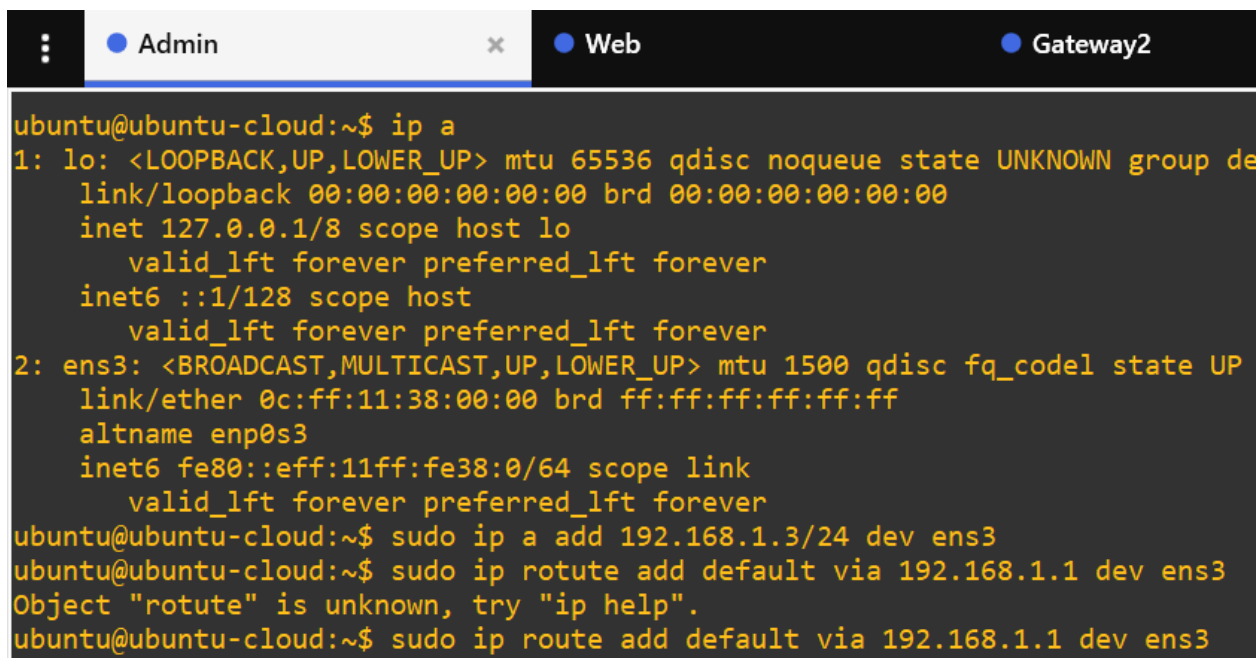
```
sudo ip a add 10.0.0.3/24 dev ens3
```

```
sudo ip route add default via 10.0.0.1 dev ens3
```

ITManager

```
sudo ip a add 10.0.0.4/24 dev ens3
```

```
sudo ip route add default via 10.0.0.1 dev ens3
```



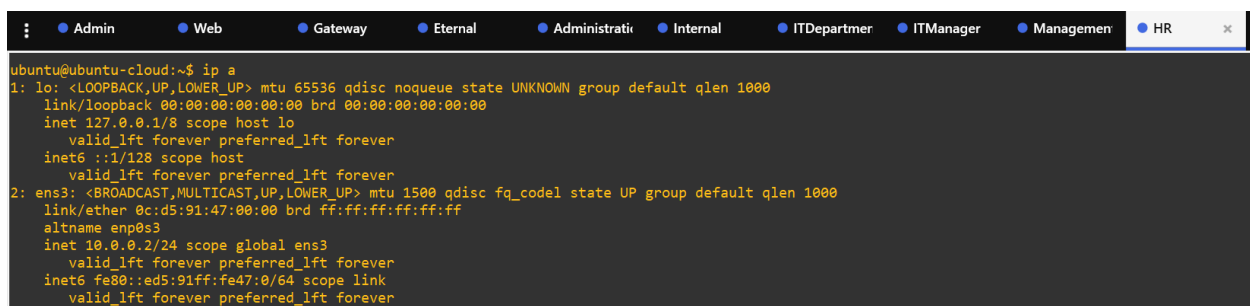
The image shows a terminal window with three tabs: 'Admin', 'Web', and 'Gateway2'. The 'Admin' tab is active. The terminal output shows the following commands and their results:

```
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 0c:ff:11:38:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet6 fe80::eff:11ff:fe38:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$ sudo ip a add 192.168.1.3/24 dev ens3
ubuntu@ubuntu-cloud:~$ sudo ip rotute add default via 192.168.1.1 dev ens3
Object "rotute" is unknown, try "ip help".
ubuntu@ubuntu-cloud:~$ sudo ip route add default via 192.168.1.1 dev ens3
```

Figure. Admin configuration

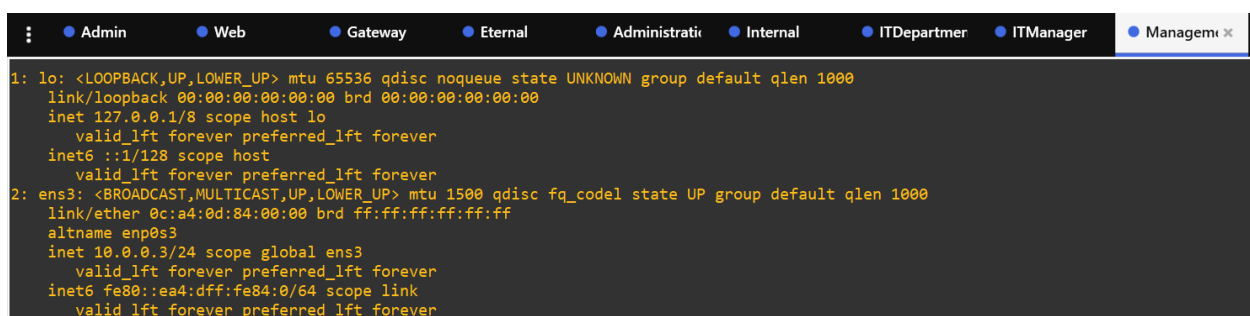
```
ubuntu@ubuntu-cloud:~$ sudo ip a add 192.168.1.2/24 dev ens3
ubuntu@ubuntu-cloud:~$ sudo ip route add default via 192.168.1.1 dev ens3
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:57:ec:d9:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.1.2/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::e57:ecff:fed9:0/64 scope link
        valid_lft forever preferred_lft forever
```

Figure. Web configuration



```
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:d5:91:47:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.0.2/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::ed5:91ff:fe47:0/64 scope link
        valid_lft forever preferred_lft forever
```

Figure. HR configuration



```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:a4:0d:84:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.0.3/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::ea4:dff:fe84:0/64 scope link
        valid_lft forever preferred_lft forever
```

Figure. Management configuration

```
Admin Web Gateway Eternal Administration Internal ITDepartment ITManager x
ubuntu@ubuntu-cloud:~$ sudo ip route add default via 10.0.0.1 dev ens3
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:ea:3d:22:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.0.4/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::eea:3dff:fe22:0/64 scope link
        valid_lft forever preferred_lft forever
```

Figure. ITManager configuration

Testing connectivity

- tested connectivity from M:M (but screenshots for ITM to all machines - for less images in the report)- success, because after configuring gateway and setting static ip address switches automatically (self-learning behaviour) assign communication based on MACs and IPs

```
Admin Web Gateway Eternal Administration Internal ITDepartment ITManager x
ubuntu@ubuntu-cloud:~$ ping 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=58.4 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=26.8 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=19.6 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=27.7 ms
^C
--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 19.620/33.117/58.390/14.921 ms
ubuntu@ubuntu-cloud:~$ ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=87.6 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=24.4 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=21.1 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=31.1 ms
^C
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 21.104/41.074/87.628/27.118 ms
ubuntu@ubuntu-cloud:~$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data.
64 bytes from 192.168.1.3: icmp_seq=1 ttl=63 time=87.7 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=63 time=34.8 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=63 time=31.4 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=63 time=50.9 ms
^C
--- 192.168.1.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 31.417/51.207/87.728/22.339 ms
ubuntu@ubuntu-cloud:~$ ping 192.168.1.4
PING 192.168.1.4 (192.168.1.4) 56(84) bytes of data.
^C
--- 192.168.1.4 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4089ms
ubuntu@ubuntu-cloud:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=63 time=68.3 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=63 time=32.4 ms
^C
--- 192.168.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 32.437/50.358/68.280/17.921 ms
```

Figure

How do VLANs work at a packet level? What are the two main protocols used for this?

VLANs operate on layer 2 (Ethernet) in a way that they split one physical switch into multiple interfaces/broadcast domains.

On access ports packets are not tagged, so they are associated only with the one VLAN that sends it — access traffic.

On the other hand, trunk port tag the packets allowing other VLANs share the same physical link (to transfer) — trunk traffic.

Protocol used:

1. 802.1Q — is a industry standard that simply inserts a 4-byte tag

| Dest MAC | Src MAC | 802.1Q TAG | EtherType | Payload | FCS |

Where frame changes (where tag is added?)

802.1Q tag inside:

TPID: 0x8100

PCP : 3 bits (priority / QoS)

DEI : 1 bit (drop eligible)

VID : 12 bits (VLAN ID, 1–4094)

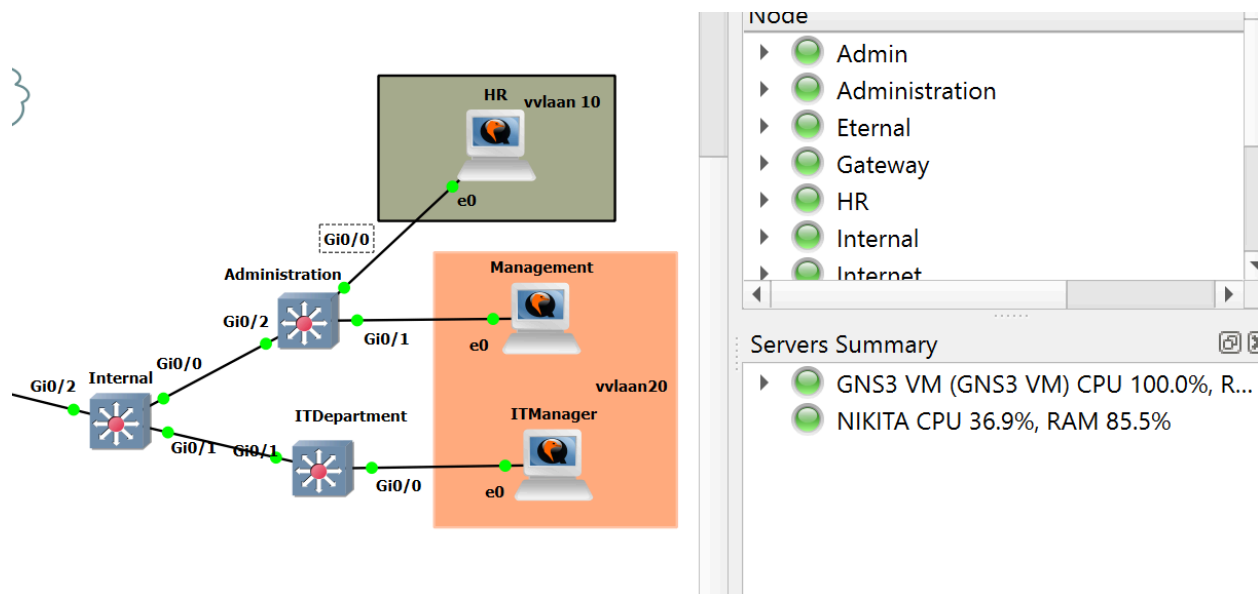
2. ISL (Inter-Switch Link) — is a proprietary protocol created, maintained and used by CISCO, his feature is to encapsulate a whole frame. But this is deprecated.

What is the Native VLAN?

Native VLAN — is a VLAN which traffic on 802.1Q trunk sends in a state “untagged” and in the same state it crosses further, without adding a tag. By default is VLAN 1 (so do not use this one number, because it is reserved — can create huge security risks).

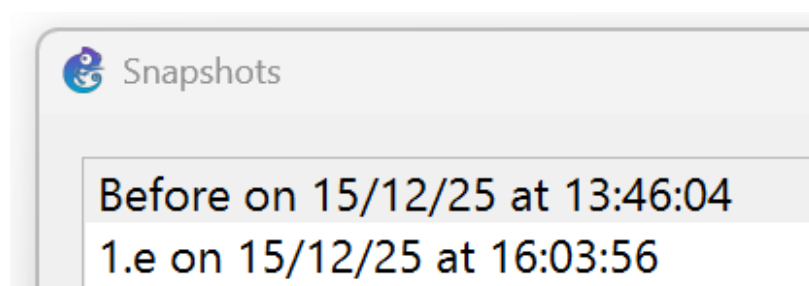
Configure the VLANs on the switches to isolate the two virtual networks as follows

- Draw figures



Figure

- Created a snapshot



- Changed configuration for nodes (firstly delete prev ip configuration, then setup new, because I did snapshot my nodes rebooted and all configs cleared, so I do not need to delete prev ip configurations):

```
# HR
sudo ip a del 10.0.0.2/24 dev ens3
sudo ip a add 10.0.10.2/24 dev ens3
sudo ip route add default via 10.0.10.1 dev ens3
```

Management

```
sudo ip a del 10.0.0.3/24 dev ens3
```

```
sudo ip a add 10.0.20.2/24 dev ens3
```

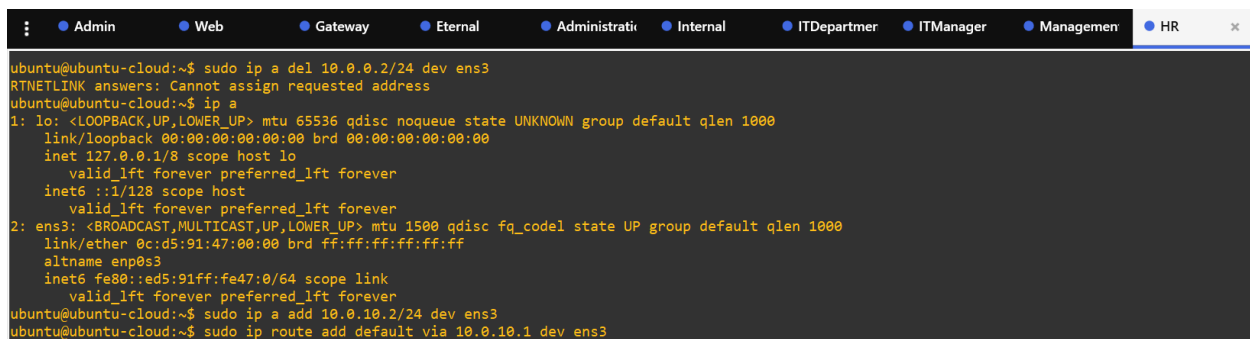
```
sudo ip route add default via 10.0.20.1 dev ens3
```

ITManager

```
sudo ip a del 10.0.0.4/24 dev ens3
```

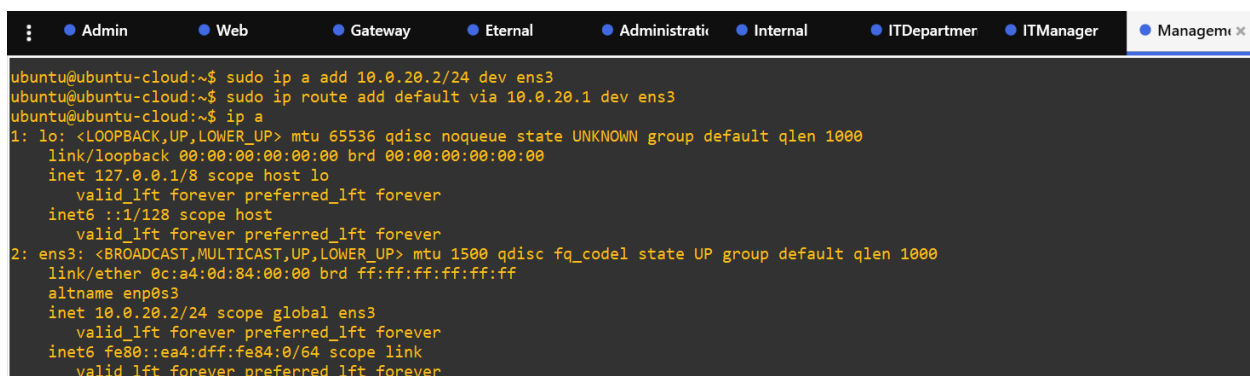
```
sudo ip a add 10.0.20.3/24 dev ens3
```

```
sudo ip route add default via 10.0.20.1 dev ens3
```



```
ubuntu@ubuntu-cloud:~$ sudo ip a del 10.0.0.2/24 dev ens3
RTNETLINK answers: Cannot assign requested address
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:d5:91:47:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet6 fe80::ed5:91ff:fe47:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$ sudo ip a add 10.0.10.2/24 dev ens3
ubuntu@ubuntu-cloud:~$ sudo ip route add default via 10.0.10.1 dev ens3
```

Figure HR



```
ubuntu@ubuntu-cloud:~$ sudo ip a add 10.0.20.2/24 dev ens3
ubuntu@ubuntu-cloud:~$ sudo ip route add default via 10.0.20.1 dev ens3
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:a4:0d:84:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.20.2/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::ea4:dff:fe84:0/64 scope link
        valid_lft forever preferred_lft forever
```

Figure Management

```

ubuntu@ubuntu-cloud:~$ sudo ip a add 10.0.20.3/24 dev ens3
ubuntu@ubuntu-cloud:~$ sudo ip route add default via 10.0.20.1 dev ens3
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:ea:3d:22:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.20.3/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::eea:3dff:fe22:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$

```

Figure ITManager

- Changed configuration for switches, so they allow VLAN, access to ports and trunk ports

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface g0/0
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface g2/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk

```

Figure Administration sw

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface g0/0
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch(config-if)#interface g0/1
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#

```

Figure ITDepartment sw

- Inspection commands on each switch executed.

```

Switch#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Gi0/2, Gi0/3, Gi1/0, Gi1/1
    Gi1/2, Gi1/3, Gi2/0, Gi2/1
    Gi2/2, Gi2/3, Gi3/0, Gi3/1
    Gi3/2, Gi3/3
20   VLAN0020               active    Gi0/0
1002 fddi-default          act/unsup
1003 token-ring-default   act/unsup
1004 fddinet-default      act/unsup
1005 trnet-default        act/unsup

```

```

Switch#
*Dec 15 13:19:29.889: %PMP-6-PMP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Gi0/1     on            802.1q         trunking      1
Port      Vlans allowed on trunk
Gi0/1     1-4094
Port      Vlans allowed and active in management domain
Gi0/1     1,20
Port      Vlans in spanning tree forwarding state and not pruned
Gi0/1     1,20

```

```

*Dec 15 13:19:33.493: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan br
VLAN Name                Status    Ports
-----
1    default                active    Gi0/2, Gi0/3, Gi1/0, Gi1/1
    Gi1/2, Gi1/3, Gi2/0, Gi2/1
    Gi2/2, Gi2/3, Gi3/0, Gi3/1
    Gi3/2, Gi3/3
10   VLAN0010               active    Gi0/0
20   VLAN0020               active    Gi0/1
1002 fddi-default          act/unsup
1003 token-ring-default   act/unsup
1004 fddinet-default      act/unsup
1005 trnet-default        act/unsup

```

```

Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Gi2/0     on            802.1q         trunking      1
Port      Vlans allowed on trunk
Gi2/0     1-4094
Port      Vlans allowed and active in management domain
Gi2/0     1,10,20
Port      Vlans in spanning tree forwarding state and not pruned
Gi2/0     1,10,20

```

Figures. vlans and trunks

Ping between ITManager and HR, do you have replies? Ping between ITManager and Management, do you have replies? Can you see the VLAN ID in Wireshark?

- Ping 1 (ITManager → HR) is not possible because they are in different subnets

```

Admin Web Gateway Eternal Administrati
ubuntu@ubuntu-cloud:~$ ping 10.0.10.2
PING 10.0.10.2 (10.0.10.2) 56(84) bytes of data.
From 10.0.20.2 icmp_seq=1 Destination Host Unreachable
From 10.0.20.2 icmp_seq=2 Destination Host Unreachable
From 10.0.20.2 icmp_seq=3 Destination Host Unreachable
^C
--- 10.0.10.2 ping statistics ---
5 packets transmitted, 0 received, +3 errors, 100% packet loss, time 4061ms
pipe 4

```

Figure

- Ping 2 (ITManager → Management) succeeded because they are in the same subnet

```

Admin Web Gateway Eternal Internal Administrati ITDepartmer ITManager x
ubuntu@ubuntu-cloud:~$ ping 10.0.20.3
PING 10.0.20.3 (10.0.20.3) 56(84) bytes of data.
64 bytes from 10.0.20.3: icmp_seq=1 ttl=64 time=0.030 ms
64 bytes from 10.0.20.3: icmp_seq=2 ttl=64 time=0.141 ms
64 bytes from 10.0.20.3: icmp_seq=3 ttl=64 time=0.187 ms
^C
--- 10.0.20.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.030/0.119/0.187/0.065 ms

```

Figure

- Wireshark shows ping traffic and VLAN ID.
 - **Access port** → frame arrives **without 802.1Q tag**
 - **Native VLAN** → frame is **explicitly untagged**
 - **Trunk (non-native VLAN)** → tag is present and visible

169 58.297482	0c:6c:c7:c1:00:00	PVST+	STP	68 Conf. Root = 32768/10/0c:6c:c7:c1:00:00
170 51.175658	10.0.20.2	10.0.20.3	ICMP	102 Echo (ping) request id=0xi
171 51.180443	10.0.20.3	10.0.20.2	ICMP	102 Echo (ping) reply id=0xi
172 52.179510	10.0.20.2	10.0.20.3	ICMP	102 Echo (ping) request id=0xi
173 52.183742	10.0.20.3	10.0.20.2	ICMP	102 Echo (ping) reply id=0xi
174 52.193712	0c:f0:87:19:00:01	PVST+	STP	68 Conf. Root = 32768/20/0c:f0:87:19:00:01
175 52.304835	0c:6c:c7:c1:00:00	PVST+	STP	68 Conf. Root = 32768/10/0c:6c:c7:c1:00:00


```

Frame 59: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
Ethernet II, Src: 0c:4e:5f:f2:00:00 (0c:4e:5f:f2:00:00), Dst: 0c:e4:b4:d2:00:00 (0c:e4:b4:d2:00:00)
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
... 0000 0001 0100 = ID: 20
Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.0.20.3, Dst: 10.0.20.2
Internet Control Message Protocol

```

Figure

Configure Inter-VLAN Routing between Management VLAN and HR VLAN and Show that you can now ping between them on gateway router

```
Gateway#enable
Gateway#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Gateway(config)#interface g2/0
Gateway(config-if)#ip address 10.0.1.1 255.255.255.0
Gateway(config-if)#no shut
```

Figure

```
Gateway(config)#interface g2/0.1
Gateway(config-subif)#encapsulation dot1Q 10
Gateway(config-subif)#ip address 10.0.10.1 255.255.255.0
Gateway(config-subif)#no shut
```

Figure

```
Gateway(config-subif)#interface g2/0.2
Gateway(config-subif)#encapsulation dot1Q 20
Gateway(config-subif)#ip address 10.0.20.1 255.255.255.0
Gateway(config-subif)#no shut
```

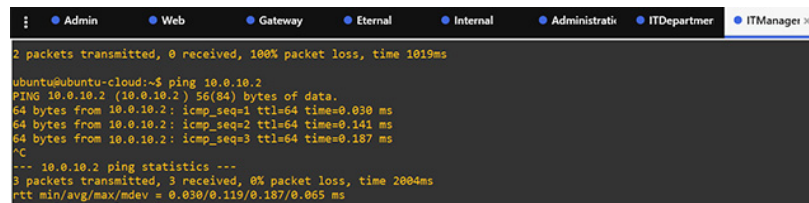
Figure

- Checked interfaces

```
Gateway#show ip inter br
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.203.131 YES DHCP    up          up
GigabitEthernet1/0       192.168.1.1     YES manual  up          up
GigabitEthernet2/0       10.0.1.1        YES manual  up          up
GigabitEthernet2/0.1     10.0.10.1       YES manual  up          up
GigabitEthernet2/0.2     10.0.20.1       YES manual  up          up
GigabitEthernet4/0       unassigned      YES unset   administratively down down
```

Figure

- Now ping succussed between itmanager→hr



```
2 packets transmitted, 0 received, 100% packet loss, time 1019ms
ubuntu@ubuntu-cloud:~$ ping 10.0.10.2
PING 10.0.10.2 (10.0.10.2) 56(84) bytes of data:
64 bytes from 10.0.10.2: icmp_seq=1 ttl=64 time=0.030 ms
64 bytes from 10.0.10.2: icmp_seq=2 ttl=64 time=0.141 ms
64 bytes from 10.0.10.2: icmp_seq=3 ttl=64 time=0.187 ms
^C
--- 10.0.10.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.030/0.119/0.187/0.065 ms
```

Figure

Fault Tolerance

What is Link Aggregation? How does it work (briefly)? What are the possible configuration modes?

Link aggregation combines/bonds multiple physical links and connection into the one logical link, using smart negotiation techniques (to choose which links are prioritized) and providing following:

bandwidth increases for logical link.

redundancy: if a link fails then, then automatically remaingin ones are set, so that connection is not interrupted

5 × 1 Gbps links → 1 logical 5 Gbps link

How it works?

Multiple connections are bonded → Traffic is being distributed using hash for MAC/IP/TCP-UDP during negotiation process → if a link in a group fails, then traffic continues on active ones.

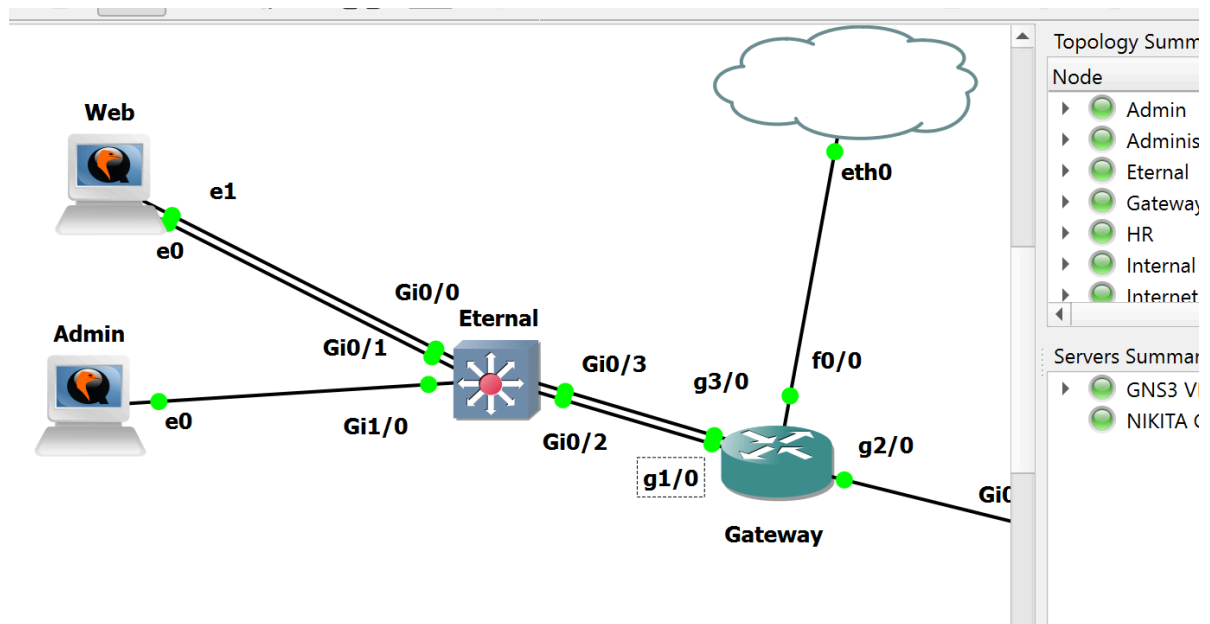
2+1 modes: static, LACP, PAgP (Port Aggregation Protocol) *.

Static — is a force-on, should be ocnfigured very precisely and accurately, otherwise risk enhances and loops can be caused.

LCAP — simply sends LACP packets, passive mode (listens only), mostly used.

Use link aggregation between the Web and the Gateway to have Load Balancing and Fault Tolerance as follows

- Infrastructure is changed:



Figure

- Cisco's 7200 included here to support for EtherChannel needed.
- Added additional network interfaces on the host and then I duplicated connections
- Then, I changed Host (Web) configuration, created a file: `sudo nano /etc/netplan/00-installer-config.yaml` and added the following netplan configuration, that created a bond between interfaceces.

```
⋮ Web x Admin
GNU nano 6.2 /etc/
network:
  version: 2
  renderer: networkd
  bonds:
    bond0:
      interfaces:
        - ens3
        - ens4
      parameters:
        mode: 802.3ad #to ens
      addresses:
        - 192.168.1.2/24
      routes:
        - to: default
          via: 192.168.1.1
  ethernets:
    ens3: {}
    ens4: {}

ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP>
    link/loopback 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host
        valid_lft forever preferred_lft 0
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft 0
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP>
    link/ether 0c:57:ec:d9:00:00
    altname enp0s3
    inet6 fe80::e57:ecff:fed9:0000/64 scope link
        valid_lft forever preferred_lft 0
3: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP>
    link/ether 0c:57:ec:d9:00:00
    altname enp0s4
```

Figure. Netplan config for web

- Router configuration

```
Gateway(config)#interface port-channel 2
Gateway(config-if)#ip address
*Dec 15 14:29:51.535: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down
Gateway(config-if)#ip address 192.168.1.1 255.255.255.0
% 192.168.1.0 overlaps with GigabitEthernet1/0
Gateway(config-if)#no shut
Gateway(config-if)#interface range g1/0 , g3/0
Gateway(config-if-range)#no ip address
Gateway(config-if-range)#channel-group 2

%Interface MTU set to channel-group MTU 1500.

GigabitEthernet1/0 added as member-1 to port-channel2

%Interface MTU set to channel-group MTU 1500.

%Interface MTU set to channel-group MTU 1500.
Gateway(config-if-range)#no shut
*Dec 15 14:30:42.511: %LINK-3-UPDOWN: Interface Port-channel2, changed state to up
*Dec 15 14:30:43.511: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up
Gateway(config-if-range)#no shut
Gateway(config-if-range)#
%Interface MTU set to channel-group MTU 1500.

GigabitEthernet3/0 added as member-2 to port-channel2

*Dec 15 14:30:48.027: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Dec 15 14:30:49.027: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
Gateway(config-if-range)#
```

Figure

- Inspection command:

```

Gateway#show inter port-channel 2
Port-channel2 is up, line protocol is up
  Hardware is GEChannel, address is ca01.080d.001c (bia ca01.080d.0000)
  MTU 1500 bytes, BW 2000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  ARP type: ARPA, ARP Timeout 04:00:00
    No. of active members in this channel: 2
      Member 0 : GigabitEthernet1/0 , Full-duplex, 1000Mb/s
      Member 1 : GigabitEthernet3/0 , Full-duplex, 1000Mb/s
    No. of Non-active members in this channel: 0
    No. of PF_JUMBO supported members in this channel : 0
  Last input 00:00:00, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/150/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/80 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    50 packets input, 3164 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
--More--

```

Figure

- Now I configure the switch (LACP mode with host-connected interfaces; static mode with router-connected interfaces because they do not support LACP)

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range g0/0 - 1
Switch(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1

Switch(config-if-range)#no shut
Switch(config-if-range)#
*Dec 15 14:33:05.103: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
*Dec 15 14:33:06.105: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
Switch(config-if-range)#channel-group 2 mode on
Creating a port-channel interface Port-channel 2

Switch(config-if-range)#no shut
*Dec 15 14:33:15.738: %LINK-3-UPDOWN: Interface Port-channel2, changed state to up
*Dec 15 14:33:16.738: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up
Switch(config-if-range)#
Switch#
*Dec 15 14:33:20.643: %SYS-5-CONFIG_I: Configured from console by console
show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        N - not in use, no aggregation
       f - failed to allocate aggregator

       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

       A - formed by Auto LAG

Number of channel-groups in use: 2
Number of aggregators:          2

```

Group	Port-channel	Protocol	Ports	
1	Po1(SU)	LACP	Gi0/0(P)	Gi0/1(P)
2	Po2(SU)	-	Gi0/2(P)	Gi0/3(P)

Figure

Test the Fault Tolerance by stopping one of the cables and see if you have any downtime.

- I paused 2 cables from GNS3 in the middle on huge ping command started on the router. But fallback was still good, operation succeeded.

```

Gateway#ping 192.168.1.2 repeat 500
Type escape sequence to abort.
Sending 500, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
.....

```

Figure



Figure

Change the topology as follows and Disable STP on the Internal network switches.

Command is used: `no spanning-tree vlan 1-4094`

Send a broadcast ping request to the PCs connected to the Internal Network.

Command is used from the gateway router: `ping 10.0.10.255`

What can you notice? Why did this happen? What are the implications of this on the network?

Loop causes 100% CPU load, STP block vanished blocking mode on my switchies, so they only in an active always forwarding state, that will lead to overload and crush of the network.

Enable back STP on the Switches and do the experiment again.

Loop is prevented.

Commands are used:

```
spanning-tree vlan 1-4094
spanning-tree mode rapid-pvst
```

Can you see STP traffic? Explain it briefly

Traffic is seen

```
-----
  Originating VLAN (PVID): 10
    Type: Originating VLAN (0x0000)
    Length: 2
    Originating VLAN: 10
```

The process:

1. Exchanging data via Bridge Protocol Data Units such as bridge ID, MAC → root (one that will have all interfaces in a forwarding state) elects
2. Non-roots select root port → remaining link chooses designated port
3. Blocking state is switched for all non-root ports and non-chosen ports

Configure the switches to have the *Internal* as the Root switch.

It is done by lowering the priority from default value as 32769 to < :

```
spanning-tree vlan 1 priority VALUE
```

Would we need STP between routers?

No. The STP disable process, which immediately caused a Layer-2 loop and a broadcast storm when a broadcast ping was sent. Re-enabling STP restored normal network behavior by blocking redundant paths.

Routers on Layer 3 so they do not forward any broadcast communication.